

Roque Enrique López Condori

Curriculum Vitae

Personal Information

Name Roque Enrique López Condori
Nationality Peruvian
Website <https://roquelopez.com>
Github <https://github.com/roquelopez>
LinkedIn <https://linkedin.com/in/roquelopez>
G. Scholar <https://scholar.google.com/citations?user=eCBv1NMAAAAJ>

Education

2013–2015 **MSc in Computer Science**, *São Paulo University*, Brazil.
2006–2010 **BSc in Computer Science**, *San Agustin National University*, Peru.

Awards

- 2023, Distinguished Participant, External Operations Platform Data Challenge (JIATF), USA.
- 2023, Best Paper Honorable Mention Award, IEEE Visualization Conference (VIS), Australia.
- 2023, Best Paper Award, Data Management for End-to-End Machine Learning (DEEM) Workshop, co-located with SIGMOD, USA.
- 2020, Member of finalist team, Wells Fargo Campus Analytics Challenge, USA.
- 2016, Finalist, Best MSc Dissertation Award, IEEE Latin American Conference on Computational Intelligence, Colombia.
- 2014, Young Scientist Best Paper Award, VII International Conference on Intelligent Information and Engineering Systems, Poland.
- 2014, Samsung Brazil Master's Scholarship (2014–2015), Brazil.
- 2013, Top 5 Bachelor's Thesis, Simposio Peruano de Inteligencia Artificial, Peru.
- 2010, Graduated in the top 5 of the undergraduate program, Peru.
- 2005, First Place and Scholarship, High School Graduation, Peru.

Experience

- Dec 2018 – **Research Engineer**, *New York University*, USA.
Present Conduct research and software engineering at NYU's Visualization, Imaging, and Data Analytics (VIDA) Lab on multiple DARPA and ARPA-H funded projects.
- Core contributor to BDI-Kit, an open-source Python toolkit for biomedical data integration and harmonization using Large Language Models (LLMs) and Natural Language Processing (NLP).
 - Extended AlphaD3M, an automated machine learning (AutoML) system, by adding support for multiple machine learning tasks, enabling automated model selection and end-to-end pipeline construction.
 - Investigated and implemented LLM-based reasoning methods for procedural text understanding, including error detection and extraction of action preconditions and effects.
- Feb 2017 – **Research Engineer**, *Inria*, France.
Aug 2018 Conducted research on extracting symbolic knowledge from unstructured text using natural language processing, machine learning, machine reading, ontologies, and Semantic Web technologies.
- Oct 2016 – **CTO**, *DoctorCV*.
Sept 2018 Co-led the technical development of a startup that uses NLP techniques to evaluate résumés and recommend improvements to increase interview success rates.
- Jul 2015 – **Researcher, Software Engineer**, *Elabora Consultoria*, Brazil.
Jan 2017 Researched information retrieval methods and word embedding techniques to automatically construct patent term hierarchies. Developed scalable solutions using Python, Apache Spark, and HBase.
- Feb 2014 – **Teaching Assistant**, *University of São Paulo*, Brazil.
Jul 2014 Teaching assistant for the course *Professional Information in Computer Science*, taught by Prof. Thiago Pardo. Assisted with grading assignments, leading discussion sessions, holding office hours, and supporting approximately 50 students.
- Oct 2013 – **Software Engineer**, *Dicionário Criativo*, Brazil.
Dec 2013 Developed a Python-based Twitter crawler to collect public discussions about books, music, and films. Built text normalization modules, including spell correction, to improve data quality.
- Jan 2012 – **Researcher, Software Engineer**, *Lindexa – Wayra Telefónica*, Peru.
Dec 2012 Researched and developed opinion mining and sentiment analysis techniques for Spanish. Implemented Python crawlers for Facebook and Twitter to collect large-scale public social media data. Lindexa was one of the winning startups in the first Wayra Peru accelerator program.
- Jan 2011 – **Technical Researcher, Developer**, *Cátedra CONCYTEC*, Peru.
Dec 2011 Conducted research on multi-document extractive summarization, text classification, and document clustering. Developed algorithms for selecting representative and informative content from large text collections.

Research Projects

- Mar 2024 – **BDF: Biomedical Data Fabric Toolbox**, *New York University*, USA.
Present ARPA-H-funded project developing methods and tools to streamline the integration and harmonization of biomedical research data from thousands of heterogeneous sources.
- Dec 2021 – **PTG: Perceptually-enabled Task Guidance**, *New York University*, USA.
Jun 2024 DARPA-funded project focused on developing AI assistants capable of providing just-in-time visual and audio guidance to support task execution.
- Dec 2018 – **D3M: Data-Driven Discovery of Models**, *New York University*, USA.
Mar 2024 DARPA-funded project focused on developing automated machine learning systems that enable domain experts to build predictive models without requiring data science expertise.

- Feb 2017 – ***ALOOF: Autonomous Learning of the Meaning of Objects***, Inria, France.
 Aug 2018 Research project focused on enabling robots and autonomous systems to learn object semantics from unstructured text, ontologies, and Semantic Web resources.
- Jul 2015 – ***Patent Faceted Search Using Topic Modeling***, Elabora Consultoria, Brazil.
 Feb 2017 Research project focused on developing information retrieval methods for faceted search and exploration of large-scale patent collections.
- Jan 2014 – ***Semantic Text Processing in Brazilian Portuguese***, Samsung – ICMC/USP, Brazil.
 Jun 2015 Research project focused on advancing semantic processing of Brazilian Portuguese, with applications to opinion mining and automatic text summarization.
- Jan 2011 – ***Automatic Summarization of Medical Records Using Natural Language Processing Techniques***, Cátedra CONCYTEC, Peru.
 Dec 2011 Research project focused on investigating automatic multi-document summarization methods for medical records using natural language processing.

Monographs

- MSc Thesis *Title:* Automatic Aspect-Based Opinion Summarization Methods
Description: This master's thesis investigates extractive and abstractive opinion summarization using an aspect-based approach. In addition to applying existing methods, two new approaches for Portuguese were proposed, achieving the best performance in the experiments.
- BSc Thesis *Title:* Medical Document Classification Based on Keywords Using Semantic Information
Description: This thesis presents a method for classifying medical documents that improves over existing approaches. The method combines statistical information with semantic relatedness between document keywords.

Languages

Spanish	Fluent	<i>native language</i>
English	Advanced Level	<i>completed studies</i>
Portuguese	Advanced Level	<i>completed studies</i>

Programming Knowledge

ML	PyTorch, Scikit-learn, Numpy, Pandas
GenAI	LangChain, MCP, LiteLLM
NLP	Transformers, NLTK, SpaCy
RL	Gymnasium, Ray
Languages	Python, Java, C++
Others	Git, Docker, Linux

Main Interests

- Data Integration
- Data-centric AI
- Applied Machine Learning
- Natural Language Processing
- Reinforcement Learning
- Artificial Intelligence